

Quiz 09

Instructions: Answer the questions with precise definitions. Your definitions should either be stated exactly as in the lecture or textbook, or be clearly equivalent to them. Having the “right idea” but not a precise formulation will not receive credit.

Problem 1. Let $c \in \mathbb{R}$ be fixed. What does it mean for a sequence of random variables (X_n) to **converge in probability** to c ?

Problem 2. Let $\hat{\theta}$ be an estimator of some target parameter $\theta \in \mathbb{R}$. Define $\text{MSE}(\hat{\theta})$.